HCI 271: CAPSTONE I

Deep Dive into Analysis & Synthesis

From Raw Data to Rigorous Insights

WEEK 05 SESSION

THE INSIGHT FUNNEL

Not Everything You Collect Is an Insight

1. OBSERVATION

Raw data. What you saw or heard. No judgment.

"P3 clicked the 'Back' button 4 times during checkout."

2. INTERPRETATION

Pattern recognition. A hypothesis about "Why".

"Users are confused by the checkout flow's step structure."

3. INSIGHT

Actionable truth. Implies a design direction.

"Users need a progress indicator to feel confident enough to proceed."

An observation describes. An interpretation explains. An insight implies action.

THE ANALYSIS PIPELINE

Raw Data Is Not Insight — Analysis Is the Bridge

RAW DATA

The "What"

Audio recordings, messy notes, photos, transcripts.

CODING

The "Tag"

Labeling specific snippets with descriptive keywords.

CLUSTERING

The "Pattern"

Grouping related codes to find commonalities.

INSIGHT

The "Why"

Actionable truths that drive design decisions.

> **"Analysis is not a summary of what happened. It is a synthesis of what it means."**

PITFALLS

The Most Common Analysis Mistake: Skipping the Funnel

THE MISTAKE

Jumping straight from raw data to design solutions.

Skipping the interpretation and validation steps.

Treating a single observation as a universal truth.

WHY IT HAPPENS

Time Pressure: "We need to ship features now."

Confirmation Bias: Seeing only what you want to see.

Lazy Thinking: Avoiding the hard work of synthesis.

THE FIX

Use the **"5 Whys"** to find root causes.

Never cite a single participant as a trend.

Require **3 independent data points** (Triangulation) before calling it an insight.

"A single quote is a clue. Three quotes from different participants is a finding."

— STEVE PORTIGAL, INTERVIEWING USERS (2023)

SESSION ROADMAP

Data to Analysis

01

Data Landscape

Attitudinal vs. Behavioral (Say vs. Do)

02

The Insight Funnel

Observation → Interpretation → Insight

03

Methods Deep Dive

Affinity Diagrams & Thematic Analysis

04

Validity

Triangulation & Verification

05

Collaboration

Managing Data as a Team

Goal: By the end of this session, you will be able to distinguish a raw data point from a rigorous, validated insight.

METHOD DEEP DIVE

Affinity Diagramming: The Full Process

01

DECONSTRUCT

Break data into atomic units. One idea, quote, or observation per note. No summaries.

02

EXTERNALIZE

Put everything on a shared wall (physical or digital). Make the invisible visible.

03

SORT

The Silent Sort. Group related notes without speaking to prevent groupthink.

04

LABEL

Name clusters with **insight statements** (full sentences), not just topic labels.

05

HIERARCHIZE

Group clusters into super-clusters to reveal higher-order themes and structure.

Source: Beyer, H., & Holtzblatt, K. (1997). Contextual Design.

ANALYSIS STEP 1

Open Coding: Stay Close to the Data

What is it?

The first pass of analysis. We read the transcript line-by-line and assign descriptive labels to specific snippets, without trying to group them yet.

TECHNIQUE: IN VIVO CODING

Use the participant's exact words as the code label.

Why? It prevents you from imposing your own bias or "solutioning" too early.

RAW TRANSCRIPT (P4)

"I tried to find the search button, but it was hidden behind the menu, so I just gave up and closed the tab."

↓

HIDDEN UI

"GAVE UP" (ABANDONMENT)

ANALYSIS PHASE 2

Axial Coding: Finding the Pattern Behind the Pattern

1. OPEN CODES

"Can't find the search bar"

"Font is too light to read"

"I got lost in the menu"

"Buttons don't look clickable"

2. CATEGORIES

Navigation Issues
Users struggle to move through the app structure.

Visual Accessibility
Elements lack sufficient contrast or affordance.

3. CORE THEME

"The interface prioritizes modern aesthetics over functional usability."

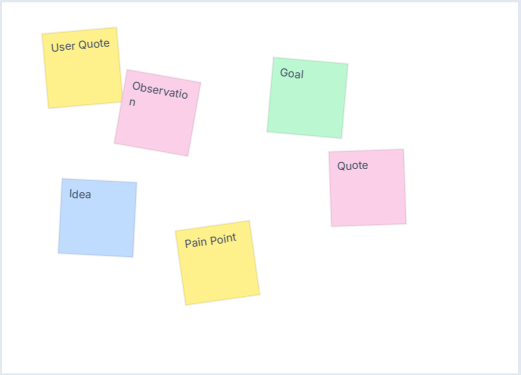
DEFINITION: Axial coding is the process of relating codes to each other to reveal causes, contexts, and consequences.

THE KJ METHOD

Affinity Diagrams: Making Patterns Visible

STEP 01

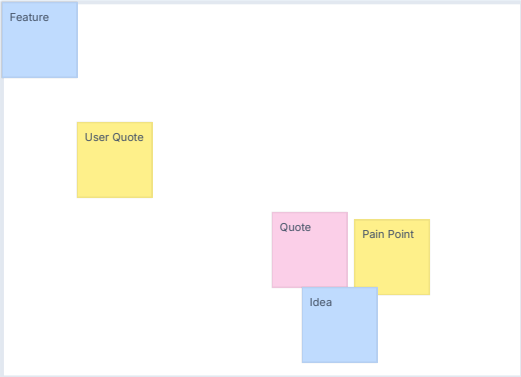
The Scramble



Deconstruct data into individual points. One thought per note. Total chaos is normal.

STEP 02

The Silent Sort



Group related notes silently. Silence prevents "groupthink" and dominant voices taking over.

STEP 03

The Label

Name the cluster with a descriptive insight, not just a category (e.g., "Trust Issues" vs "Security").

Pro Tip: If a note belongs in two groups, duplicate it. Don't force a choice.

BEST PRACTICES

Affinity Diagramming: What Goes Wrong

COMMON PITFALLS

"Topic Bucketing"

Grouping notes by generic nouns like "Price" or "Usability" without saying anything about them.

Summarizing Data

Writing "User liked it" instead of the exact quote. You lose the nuance and emotion.

THE FIX

Insight Statements

Write full sentences that describe the relationship. "Users prioritize transparency over low cost."

Atomic Notes

One idea per note. Use verbatim quotes. If a note has two ideas, split it.

Source: Nielsen Norman Group, "Pitfalls of Affinity Diagramming." The goal is not to organize the data, but to *synthesize* it.

METHOD DEEP DIVE

Thematic Analysis: A Rigorous Approach

01 Familiarize
Read and re-read the data. Note initial ideas. Immerse yourself.

02 Generate Initial Codes
Code interesting features systematically across the entire data set.

03 Search for Themes
Collate codes into potential themes. Gather all data relevant to each potential theme.

04 Review Themes
Check if themes work in relation to the coded extracts (Level 1) and the entire data set (Level 2).

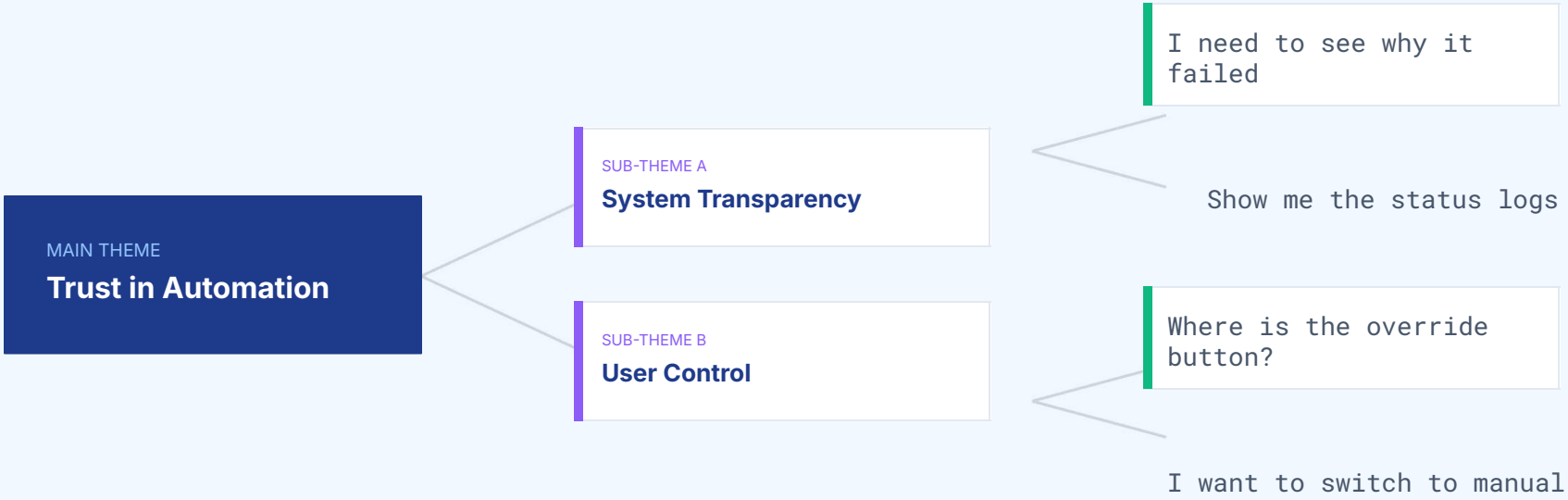
05 Define & Name
Refine the specifics of each theme. Generate clear definitions and names.

06 Produce the Report
Select vivid examples. Relate back to the research question and literature.

Source: *Braun, V. & Clarke, V. (2006). "Using thematic analysis in psychology". Qualitative Research in Psychology, 3(2), 77-101.*

VISUALIZATION

Building a Thematic Map



Codes: The raw building blocks (Quotes/Observations)

Sub-Themes: Groupings of related codes

Themes: The overarching story or argument

IN-CLASS EXERCISE

Can You Tell the Difference?

The user paused for 5 seconds before clicking 'Checkout'.

OBSERVATION

The user is hesitant because they are worried about shipping costs.

INTERPRETATION

Users need upfront transparency on costs to build trust before commitment.

INSIGHT

Discussion: Why is it dangerous to mix these up in your research report?

VALIDITY & VERIFICATION

One Source Is a Clue. Three Sources Are Evidence.

Triangulation is the use of multiple methods or data sources to develop a comprehensive understanding—it prevents you from being misled by outliers.

METHODOLOGICAL

Using different methods to study the same problem. *Example:* Combining **Survey** results (Quant) with **User Interviews** (Qual).

DATA SOURCES

Comparing data from different times, spaces, or people. *Example:* Comparing **New Users** vs. **Power Users** or Weekday vs. Weekend usage.

INVESTIGATOR

Multiple researchers analyze the same data to minimize bias. *Example:* Two designers independently coding the same interview transcripts.

If all three point to the same conclusion, you have a validated insight.

VALIDITY CHECK

How Do You Know Your Codes Are Consistent?

⚖ Inter-Rater Reliability (IRR)

The degree of agreement among independent raters. It measures whether your coding scheme is objective or just your personal opinion.

1. Two researchers code the same 10% of data independently.
2. Compare the codes applied to each segment.
3. Calculate agreement % (Aim for >80%).
4. Discuss disagreements and refine the Codebook definitions.

CODING COMPARISON EXAMPLE

Transcript Segment	Coder A	Coder B	
"I couldn't find the search bar."	NAV_ISSUE	NAV_ISSUE	✓
"It took too long to load."	PERF_LAG	UX_FRUST	✗
"The colors are too bright."	VISUAL_UI	VISUAL_UI	✓

** The mismatch in row 2 reveals a need to define if "Performance" is separate from general "Frustration".*

"If you can't agree on the code, your definition is too vague."

LOGISTICS

When the Data Gets Overwhelming

CHUNKING

Don't try to analyze 20 interviews at once. Break them down by **Persona**, **Task**, or **Feature**.

Tip: Analyze 3 interviews, build a codebook, then apply it to the rest.

SATURATION

The Stopping Rule. You stop coding when new data yields no new codes or themes.

Tip: Usually happens after 5-8 interviews per user group.

DIGITAL TOOLS

Sticky notes fail after 100 points. Move to software (Dedoose, Dovetail, NVivo, Airtable) for tagging and retrieval.

Tip: Use tools to store data, but use walls to synthesize it.

"Analysis paralysis happens when you try to hold the entire dataset in your head at once."

COLLABORATIVE SYNTHESIS

Synthesis Is a Team Sport. But It Has Rules

Assign Roles

The Facilitator

Keeps the team moving. Enforces the "Silent Sort" rule. Stops circular debates.

The Scribe

Captures the exact wording of insights. Ensures nothing is lost in translation.

The Skeptic

Challenges assumptions. Asks "Which user said that?" whenever a generalization is made.

Follow Protocols

Evidence First: No insight goes on the board without a data point (quote or observation) backing it up.

No Silent Consensus: If you disagree with a grouping, move the note. Disagreement reveals nuance.

The Parking Lot: Great ideas that aren't supported by *this* data set go here. Don't lose them, but don't let them bias the analysis.

"Collaboration is about conflict resolution, not just agreement."

SYNTHESIS TO ACTION

The Final Step: Turning Insights into Design Directions

THE INSIGHT

The provocative truth you discovered about user behavior.

EXAMPLE

Users freeze when faced with five subscription tiers.

IMPLICATION

The constraint this truth places on the design.

EXAMPLE

Show just two options up front; reveal more progressively.

HMW QUESTION

A creative prompt to guide ideation.

EXAMPLE

How might we guide users to the right plan without a full comparison?

Research doesn't tell you what to design; it tells you where to focus your creativity.

IN-CLASS ACTIVITY

Workshop: From Transcript to Insight

01

10 MINS

INDIVIDUAL CODING

Read the sample transcript silently. Highlight key phrases. Write 5-10 codes on sticky notes (one per note).

02

15 MINS

TEAM CLUSTERING

Post your notes on the wall. Group them silently. Move notes around until clusters emerge. No talking yet.

03

10 MINS

SYNTHESIS

Discuss the clusters. Name each group with an "Insight Statement" (a full sentence, not just a topic).

Deliverable: Take one clear photo of your team's final Affinity Diagram.

ACTIVITY

Workshop: From Data Dump to Design Direction

10 MIN

1. CLASSIFY

Review raw notes. Sort into two piles: **Observations** (Facts) vs. **Interpretations** (Opinions).

15 MIN

2. CODE

Tag observations. Find recurring behaviors or pain points. Create a mini-codebook (3-5 codes max).

20 MIN

3. SYNTHESIZE

Cluster your codes. Write **one Insight Statement** that explains the "Why" behind the behavior.

FINAL OUTPUT **One validated insight statement backed by at least 3 distinct data points (Triangulation).**

"The insight is already in the data. Your job is to find it."

Analysis is not about invention.
It is about rigorous discovery.

COMING UP NEXT WEEK

From Insights to Ideation & Prototyping